Exploring the use of LID in learning disentangled and cluster-friendly latent spaces in Autoencoders

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***Abstract*—How deep learning models work and make inferences in their latent space is very unclear and a subject of active research. Disentangled latent representations are one of the ways to achieve interpretability where every latent dimension encodes a distinct, meaningful factor of variation in data. Thus it is crucial to enforce such disentanglement on model learning. Another important use of latent representations is in downstream analyses tasks such as clustering. In this report, we explore the various ways we can utilize local intrinsic dimensionality of data in latent space to achieve such disentangled or cluster-friendly latent spaces.**

***Index Terms*—disentanglement, latent, dimension, correlation, intrinsic, clustering**

I. Introduction

Characterization and quantification of similarity between various models is a fertile field of research in the subject of neural network interpretability. Hence, numerous research works (1; 2; 3) emphasize the measurement of statistical similarities between the representation achieved by various models. The measurement itself serves as clarification or bridging between various facets of data representation. It also aids in determining the compatibility of latent spaces, indicating that information garnered from one model can be efficiently transferred to another. Conventional latent similarity metrics such as CKA(1) and DistanceCorrelation (4) find a very low structural correlation between paired multimodal representations because of the strongly non-linear nature of these correlations.

Present state-of-the-art deals with using intrinsic dimensionality to discover correlations between multimodal representations that have a highly non-linear relationship which is talked about in (5). In this work, the authors suggest utilizing Normalized Mutual Information as the correlation estimation technique with intrinsic dimensionality instead of entropy to find non-linear relations in the high-dimensional latent space of multimodal data. They try to discover if multimodal data, for instance, image captioning data, while being processed by different deep learning models, for example, a vision model and a language model, produce hidden representations that are correlated with each other. This allows the model to learn which caption is paired with which image.

Although this approach enables us to see how models learn to connect data in learned latent space, the representations are still an entangled ensemble of variations coming from various features of the data. Hence, the best way to increase the interpretability is to disentangle these representations and forcing the model to learn a by parts representation where one latent representation depicts a specific part of the data and is independent of the other. Say for example, when we humans try to distinguish between a Tesla Model S and a BMW Series 5, we subconsciously disentangle different attributes, analyzing them separately before forming an overall judgment - such as how Tesla has a sleek, streamlined body with slender, futuristic headlights and is quiet with electric acceleration whereas the BMW possesses a more conventional, muscular sedan appearance with aggressive-looking LED headlights and a conventional combustion engine with distinctive exhaust note. Instead of perceiving the two cars as a single indistinct entity, the brain naturally separates factors like shape, headlights and performance, allowing for a structured comparison. Latent spaces learned through models such as autoencoders provide compact and informative representations that are particularly well-suited for downstream analytical tasks, including clustering. Real-world datasets are often embedded in high-dimensional spaces characterized by noise, sparsity, and redundancy. However, these data points frequently lie on a much simpler, lower-dimensional manifold. Autoencoders—unsupervised neural architectures—are trained to compress input data into such latent representations, retaining only the most essential features while filtering out extraneous or redundant dimensions.

Through dimensionality reduction, these learned latent spaces significantly reduce the complexity of high-dimensional data. The resulting representations emphasize discriminative and meaningful features, thereby simplifying subsequent analytical tasks. Clustering in these latent spaces becomes more efficient and effective, as the reduced dimensionality mitigates issues associated with the "curse of dimensionality"—a common limitation in high-dimensional clustering that leads to poor performance and interpretability.

For instance, clustering directly on raw pixel values in image datasets is both computationally intensive and often ineffective. The MNIST dataset, for example, consists of 784-dimensional input vectors representing grayscale pixel values, the majority of which are zero. This creates a highly sparse and high-dimensional feature space. Reducing this space to a lower-dimensional latent representation (e.g., 10–12 dimensions) facilitates the identification of meaningful clusters corresponding to different digits, improving both computational efficiency and clustering accuracy.

This report investigates how disentangled and cluster-friendly latent representations can be encouraged by leveraging the concept of intrinsic dimensionality. The primary contributions of this work are as follows:

- An exploration of how local intrinsic dimensionality (LID) can be used to enforce part-wise learning in autoencoder latent spaces, via an approximation of Total Correlation.
- A study on how learning cluster-friendly latent spaces can be enhanced through the use of Itakura–Saito divergence (6), both as a replacement for and in conjunction with the traditional Kullback–Leibler divergence (6).

II. Background

Intrinsic dimensionality (7)is the minimum number of independent variables required to describe a dataset without loss of its intrinsic structure. It is the complexity of the data itself, even if it's in a very high-dimensional space. Most datasets (e.g., images or sensor data) are in high-dimensional spaces but lie on a lower-dimensional manifold. Intrinsic dimensionality helps in the removal of redundant information. Methods like PCA (8), t-SNE (9), or autoencoders (10) aim to find the intrinsic dimensionality and project data into a lower-dimensional space without losing critical information. Machine learning models often fall victim to the "curse of dimensionality"(11). Finding and working on intrinsic dimensionality improves generalization and efficiency. Several methods exist for estimating the ID of data representations. The Two-NN (Two Nearest Neighbors) estimator (12) calculates the Local Intrinsic Dimensionality (LID) by examining the ratio of distances to the two nearest neighbors of each data point. Specifically, it uses the distances to the first and second nearest neighbors to estimate how data density changes locally around each point. This ratio helps approximate the intrinsic dimension, capturing local geometric complexity and data manifold properties, particularly useful for dimensionality reduction and identifying clusters or outliers. Unlike linear dimensionality reduction techniques such as PCA, TwoNN can accurately estimate ID even when data lies on curved, topologically complex manifolds with non-uniform sampling . A convincing pattern emerges when looking at ID across the layers of trained neural networks: a standard "hunchback" shape in which ID first increases in shallow layers and then gradually decreases in deeper layers (13). This suggests that neural networks initially expand the dimensionality of representations to disentangle features before compressing information onto lower-dimensional manifolds that preserve the underlying structure for classification. This dimension reduction in later layers is strongly correlated with generalization performance. Networks that are able to reduce dimensionality well in their final layers perform better on test data, which suggests that good generalization must map inputs to low-dimensional, task-specific manifolds.

In the paper (5), the authors present the concept that if two dataset representations are correlated then the intrinsic dimensionality of the dataset created by the row-wise concatenation of the features of these representations is reduced because the information in one representation can describe some aspects of the other. Accordingly they propose $I_d Cor$, a novel correlation metric based on intrinsic dimension estimation that can unveil the nonlinear relationship between high-dimensional data manifolds. The correlation is based on normalized mutual information (NMI) metric (14). Computing mutual information in high-dimensional data is challenging due to curse of dimensionality because of which, the authors propose to use intrinsic dimension as a proxy for the information content of a dataset. In the paper (6), the authors develop substantial new theory that asymptotically relates tail entropy, divergences and LID. The concept of Local Intrinsic Dimensional Entropy proposed in (15) has been introduced, which utilizes the average local intrinsic dimension of a distribution as a robust entropy measure. This approach captures the data's structural properties and offers advantages in contexts like deep learning, where data undergoes multiple transformations.
Total Correlation (TC) (16), also known as multi-information, is a generalization of mutual information for measuring dependence among more than two random variables. It quantifies the amount of total information in a set of variables beyond what would be there if the variables were independent. TC uses entropy as a measure of information contained in a variable. Thus, it is worthwhile to explore the use of intrinsic dimensionality as a proxy for entropy in TC and come to a theoretical and practical agreement on whether this approach is actually fruitful.

Clustering is an unsupervised machine learning technique used to group similar unlabeled data points into clusters based on defined similarity measures. Cluster analysis can be applied to various domains like market segmentation, social network analysis, and medical imaging to identify patterns and simplify complex datasets. Clustering high-dimensional data poses several significant challenges due to the complexity and intrinsic properties of such data (17).

- Distances between data points become less meaningful as dimensionality increases, leading to points being nearly equidistant. This makes distinguishing clusters difficult.
- High dimensionality exponentially increases computational time and resource requirements.
- Many dimensions could be noisy or irrelevant, negatively affecting clustering performance.
- Data points in high dimensions tend to lie on a lower-dimensional manifold, causing issues with traditional clustering methods that assume a full-dimensional distribution.

Thus dimension reduction becomes a crucial step before applying clustering. Present state of the art for learning lower dimensional latent representations for high-dimensional datasets with highly non-linear relationships are autoencoders (10). An autoencoder is a neural network architecture designed for unsupervised learning that encodes input data into a lower-dimensional latent representation and then reconstructs the original data from this representation. It consists of two main parts: an encoder that compresses the data into a compact latent space, and a decoder that reconstructs the data back to its original form. Training involves minimizing the reconstruction error, thus enabling effective feature extraction, dimensionality reduction, and noise filtering. The representations learnt are often entangled and hard to interpret because it is trained only for reconstruction, not for explicit factor separation. They are also not inherently cluster-friendly as these representations aren't guaranteed to align with meaningful or well-separated clusters unless explicitly guided or regularized.

III. Proposed Methodology

A. Disentanglement using LID

We talked about how autoencoders learn representations that are often entangled and hard to interpret. One way to force the autoencoder to learn by-parts representation in the latent space is to apply a constraint. Currently, autoencoders are trained to minimize the reconstruction loss without any regard for the disentangling of learned representations. We can enforce the disentanglement using the theory of Total Correlation. The mathematical definition of Total Correlation is as follows: For a set of n random variables $X_1, X_2, ..X_n$ with a joint probability distribution $p(X_1, X_2, ..X_n)$, TC is defined as:

$$TC(X_1, X_2, ..X_n) = \sum_{i=1}^{n} H(X_i) - H(X_1, X_2, ..X_n) \quad (1)$$

where $H(X_i)$ is the Shannon entropy (18) of the individual variable X_i while $H(X_1, X_2, ..X_n)$ is the joint entropy of all variables together. Intuitively, here instead of using entropy, we can directly use the I_d of these variables as proxy. So the new expression will be:

$$TC_{I_d}(X_1, X_2, ..X_n) = \sum_{i=1}^{n} I_d(X_i) - I_d(X_1, X_2, ..X_n) \quad (2)$$

However, the correctness of this formulation is subject to question. Most notably, Total Correlation (TC) is defined over the full data distribution, whereas Local Intrinsic Dimensionality (LID) characterizes behavior in the lower distributional tail. As a result, directly substituting entropy terms in the TC formulation with LID compromises the theoretical soundness and stability of the resulting metric. But this inconsistency presents an opportunity to study the behavior of TC in the limit where the radius of the neighborhood approaches zero and, perhaps, to write down in terms of LID or revealing that such a limit may not exist. Before empirically evaluating the accuracy of Equation (2), we first undertake a theoretical investigation into the plausibility of the proposed relationship between TC and LID. The asymptotic connection between these two quantities is guided in part by the derivations presented in (6).

$$\begin{aligned}
\lim_{w->0} TC(Z_1, Z_2, ..Z_n; w) &= \lim_{w->0} \sum_{i=1}^{n} H(Z_i; w) - \\
&\quad H(Z_1, Z_2, ..Z_n; w) \\
&= \lim_{w->0} \sum_{i-1}^{n} \left[-\int_0^w Z'_{iw}(t) \ln(Z'_{iw}(t)) \right] - \\
&\quad \left[-\int_0^w Z'_{w}(t) \ln(Z'_{w}(t)) \right] \\
&= \lim_{w->0} \sum_{i-1}^{n} \left[-\int_0^w ID_{Z_i}(t) \frac{Z_i(t)}{t} \ln\left(ID_{Z_i}(t) \frac{Z_i(t)}{t}\right) dt \right] \\
&\quad - \left[-\int_0^w ID_{Z}(t) \frac{Z(t)}{t} \ln\left(ID_{Z}(t) \frac{Z(t)}{t}\right) dt \right] \\
&= \lim_{w->0} \sum_{i-1}^{n} \left[-\int_0^w \frac{ID^*_{Z_i}}{t} \left(\frac{t}{w}\right)^{ID^*_{Z_i}} \ln\left(\frac{ID^*_{Z_i}}{t} \left(\frac{t}{w}\right)^{ID^*_{Z}} \right) dt \right] \\
&\quad - \left[-\int_0^w \frac{ID^*_{Z}}{t} \left(\frac{t}{w}\right)^{ID^*_{Z}} \ln\left(\frac{ID^*_{Z}}{t} \left(\frac{t}{w}\right)^{ID^*_{Z}} \right) dt \right] \\
&= \lim_{w->0} \sum_{i-1}^{n} \left[1 - \frac{1}{ID^*_{Z_i}} - \ln \frac{ID^*_{Z_i}}{w} \right] \\
&\quad - \left[1 - \frac{1}{ID^*_{Z}} - \ln \frac{ID^*_{Z}}{w} \right]
\end{aligned} \quad (3)$$

Here $Z_i \in \{Z_1, Z_2, ..Z_L\}$ are individual latent dimensions or features and Z is the entire latent space. As we see from (3), the TC value diverges as the neighborhood radius approaches zero, implying that TC cannot be reliably expressed in terms of LID. To validate this theoretical observation, an empirical investigation was also conducted to assess whether LID-based approximations of TC could facilitate disentangled representation learning.

To this end, an autoencoder architecture was implemented with a composite loss function comprising both the reconstruction loss and the TC loss, where the latter was modeled as shown in Equation (2). The TC loss term was scaled by a small hyperparameter λ, ensuring it would not overshadow the reconstruction loss during training. Conceptually, a positive TC loss suggests the presence of shared information among latent features. Therefore, minimizing this loss aims to encourage statistical independence across dimensions, promoting the emergence of disentangled, self-contained features that individually contribute to data reconstruction.

B. Learning cluster-friendly latent space

Due to the inherent challenges associated with clustering in high-dimensional spaces, dimensionality reduction is frequently employed as a critical preprocessing step. Traditional linear techniques such as Principal Component Analysis

(PCA) or Non-negative Matrix Factorization (NMF) (19) are limited in their ability to capture complex non-linear structures within the data. Although non-linear methods such as t-SNE (9) and UMAP (20) are more adept at preserving local structures, they often fail to maintain global relationships between clusters and can be computationally intensive. In contrast, deep learning approaches like autoencoders excel at modeling intricate non-linear dependencies; however, the latent spaces they generate are not always conducive to effective clustering.

To ensure that the autoencoder learns a latent space that facilitates clustering, several modifications were introduced to the baseline architecture. The autoencoder comprises an encoder that maps the input data to a latent representation, and a decoder that reconstructs the input from this representation. During training, weighted K-means clustering (21) was applied directly to the latent space. Specifically, a set of cluster centroids was initialized based on the desired number of clusters. The Euclidean distance between each latent point z_i and centroid c_r was computed and then weighted by w_{ir}, which represents the probability of point z_i belonging to cluster c_r. The K-means loss was then defined as the mean of the product between these distances and their corresponding weights, as follows::

$$L_{km} = \frac{1}{nv} \sum_{i=1}^{n} \sum_{r=1}^{v} w_{ir} ||z_i - c_r||^2 \tag{4}$$

The weights are assigned using the softmax function such that it gives us the similarity measure between a point and it's corresponding centroid:

$$w_{ir} = \frac{\exp\left(-||z_i - c_r||^2\right)}{\sum \exp\left(-||z_i - c_r||^2\right)} \tag{5}$$

The more distant a point is from the center, the smaller weight it will be assigned.

To make sure similar points are clustered together and dissimilar points are as far apart as possible, a KL divergence loss function is introduced in models like t-SNE (9), scDeepCluster (22) and scziDesk (23). First a probability distribution of every point z_i belonging to the same cluster as other points in latent space is built using t-distribution kernel as follows:

$$p_{j|i} = \frac{(1 + ||z_i - z_j||^2/t)^{-(t+1)/2}}{\sum_{i \neq k}(1 + ||z_i - z_k||^2/t)^{-(t+1)/2}} \tag{6}$$

where t controls the degrees of freedom, essentially giving the distribution heavier tails than that of a gaussian distribution. An auxiliary target distribution is defined as follows:

$$q_{j|i} = \frac{p_{j|i}^2 \sum_{i \neq j} p_{j|i}}{\sum_{i \neq k} p_{k|i}^2 \sum_{i \neq j} p_{k|i}} \tag{7}$$

The target distribution exhibits sharper tails, wherein lower probabilities become vanishingly small while higher probabilities approach unity. The core objective is to align the distribution $p_{j|i}$ with the target distribution $q_{j|i}$, thereby compressing the heavy tails and producing more pronounced peaks. This transformation effectively brings similar data points closer together and pushes dissimilar points further apart, enhancing cluster separation in the latent space. Consequently, the Kullback–Leibler (KL) divergence loss is employed to encourage the model distribution p to approximate the target distribution q.:

$$Loss_{KL} = \sum_{i=1}^{N} q_{j|i} \log \frac{q_{j|i}}{p_{j|i}} \tag{8}$$

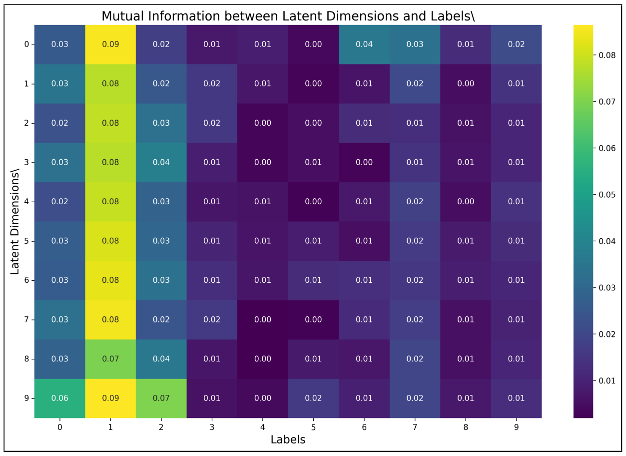
Finally, I introduce the Itakura-Saito divergence using estimated LIDs of distributions $q_{j|i}$ and $p_{j|i}$ as follows:

$$Loss_{IS} = \left(\frac{\widehat{ID_P^*}}{\widehat{ID_Q^*}} \right) - \log \left(\frac{\widehat{ID_P^*}}{\widehat{ID_Q^*}} \right) - 1 \tag{9}$$

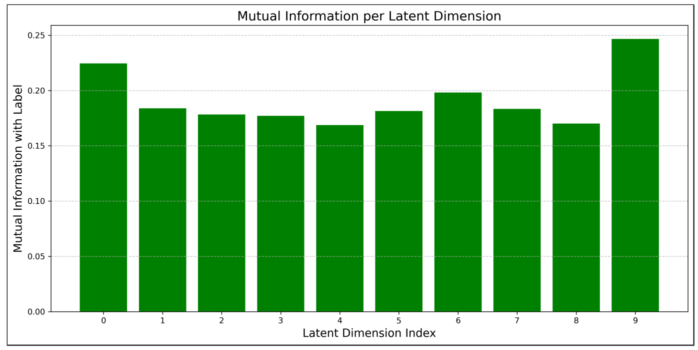
- What this loss essentially models is that, how many degrees of freedom does a point in distribution P have relative to that of a point Q. This gives manifold awareness to the clusters - Pulling toward a low (or uniform) target encourages the encoder to flatten out wrinkle-like variations inside a cluster, giving tighter, more linear latent shapes which are more separable. This is exactly what recent intrinsic-dimension regularizers (24) exploit to improve discriminability and fairness
- IS Loss emphasizes ratios instead of absolute errors. On data whose neighborhood sizes vary strongly, the KL term can over-tighten dense regions and neglect sparse ones; an LID term counter-balances that by pushing sparse zones to contract as well.
- Outliers typically have abnormally high LID. Penalising that keeps them from dragging centroids, a trick borrowed from LID-based anomaly detectors (25)
- Low LID means the data is more tightly packed and hence the there is less room for adversaries. Just optimizing P and Q leads to larger clusters swallowing the smaller ones. Keeping LID low everywhere will make sure the bigger clusters do not cover the smaller ones. Although we have these theoretical guarantees, the practical viability of this method is explored in the Results section.

C. LID estimator

We can estimate I_ds using Two-NN estimator as was discussed in (5). Briefly, the estimator operates by first computing the ratio between the distances to the second and first nearest neighbors of each data point, which is then transformed using an inverse logarithmic scale. Subsequently, it evaluates the cumulative distribution function (CDF) of these values relative to each point. A line is then fitted through the origin, using the inverse log-scale ratios as the x-coordinates and the corresponding CDF values as the y-coordinates. The slope of this line provides the estimated Local Intrinsic Dimensionality (LID) of the data. The use of only the two nearest neighbors renders the estimator highly efficient and free of tunable parameters.

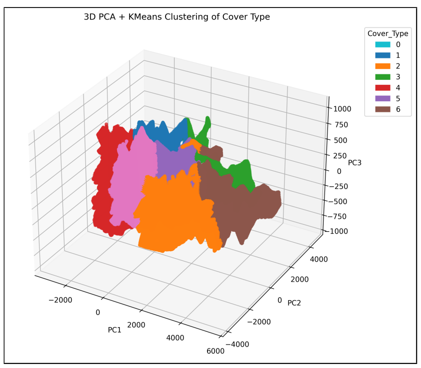


(a) Mutual Information Heatmap

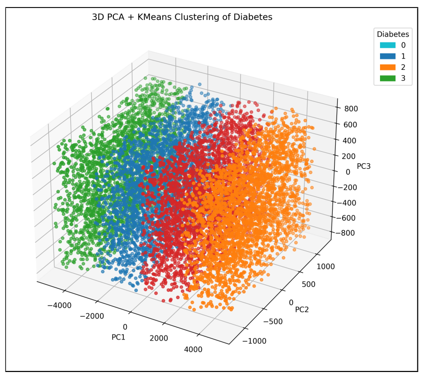


(b) Mutual Information Scores

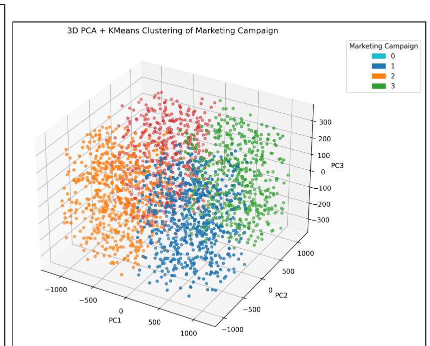
Fig. 1: The amount of disentanglement measured by (a) how a informative a particular latent dimension is in expressing a particular label and (b) how much mutual information is shared between a particular latent dimension and label ’0’



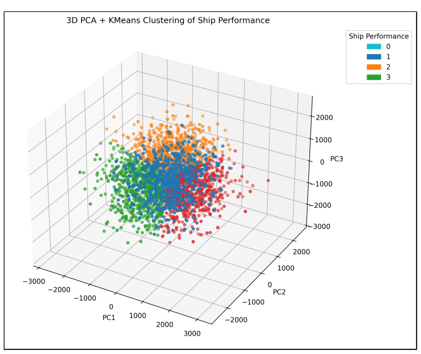
(a) Cover Type



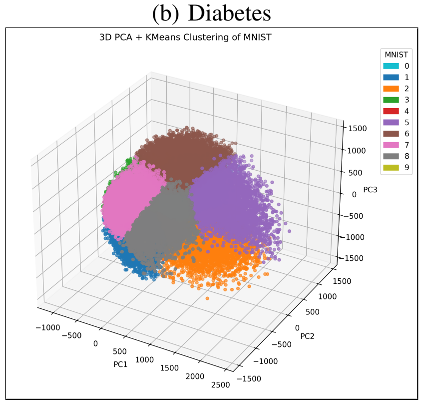
(b) Diabetes



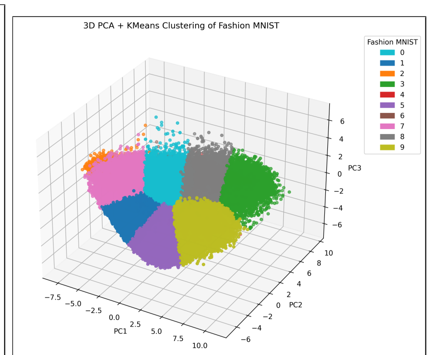
(c) Marketing Campaign



(d) Ship Performance



(e) MNIST



(f) Fashion MNIST

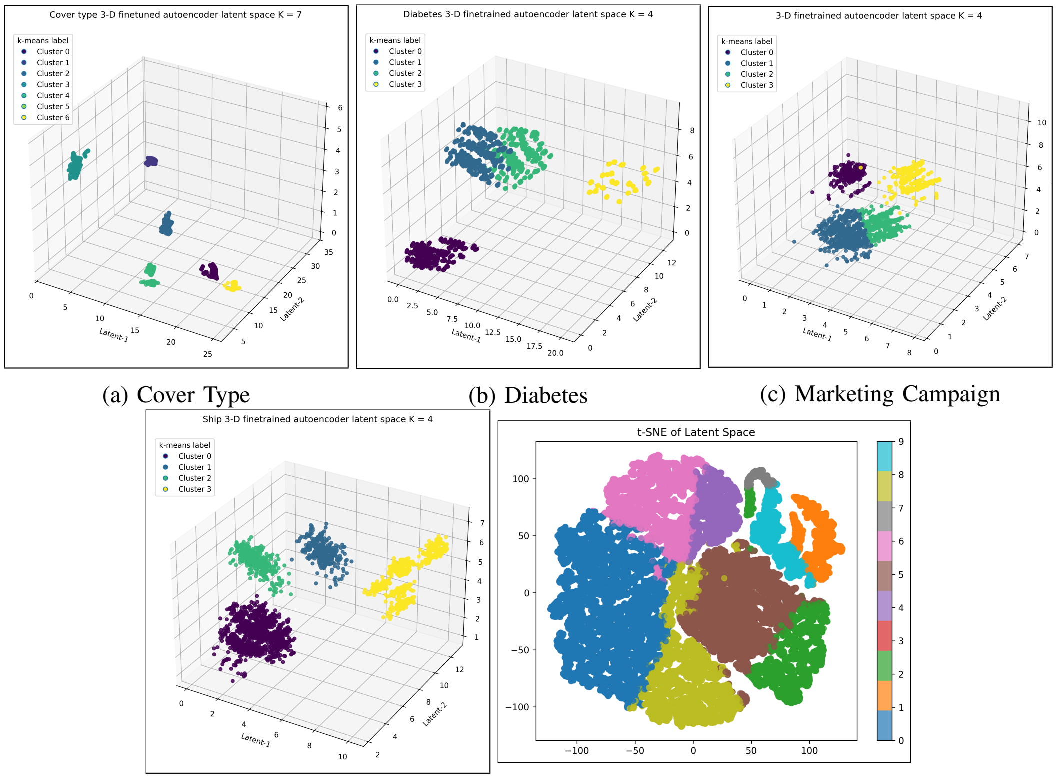
Fig. 2: PCA + Kmeans performed as a baseline

Dataset used	Samples	Features
MNIST	60000	784
Ship Performance	2736	24
Diabetes	152	20
Cover Type	581012	54
Market	2240	29
Fashion-MNIST	60000	784

TABLE I: Datasets

IV. Results

A simple autoencoder architecture was developed, consisting of an encoder and decoder each containing a single hidden layer, alongside a central bottleneck layer. Network weights were optimized using the Adam optimizer. Training was conducted utilizing an NVIDIA A100 GPU; however,



(a) Cover Type (b) Diabetes (c) Marketing Campaign

(d) Ship Performance (e) MNIST

Fig. 3: Itakura-Saito Divergence loss using estimated LID is used to replace KL divergence loss

CPU-based training is also feasible, with the exception of one PyTorch library dependency within the LID estimator module, which explicitly requires CUDA support. The datasets employed for experimentation are presented in Table I

A. Disentanglement using LID

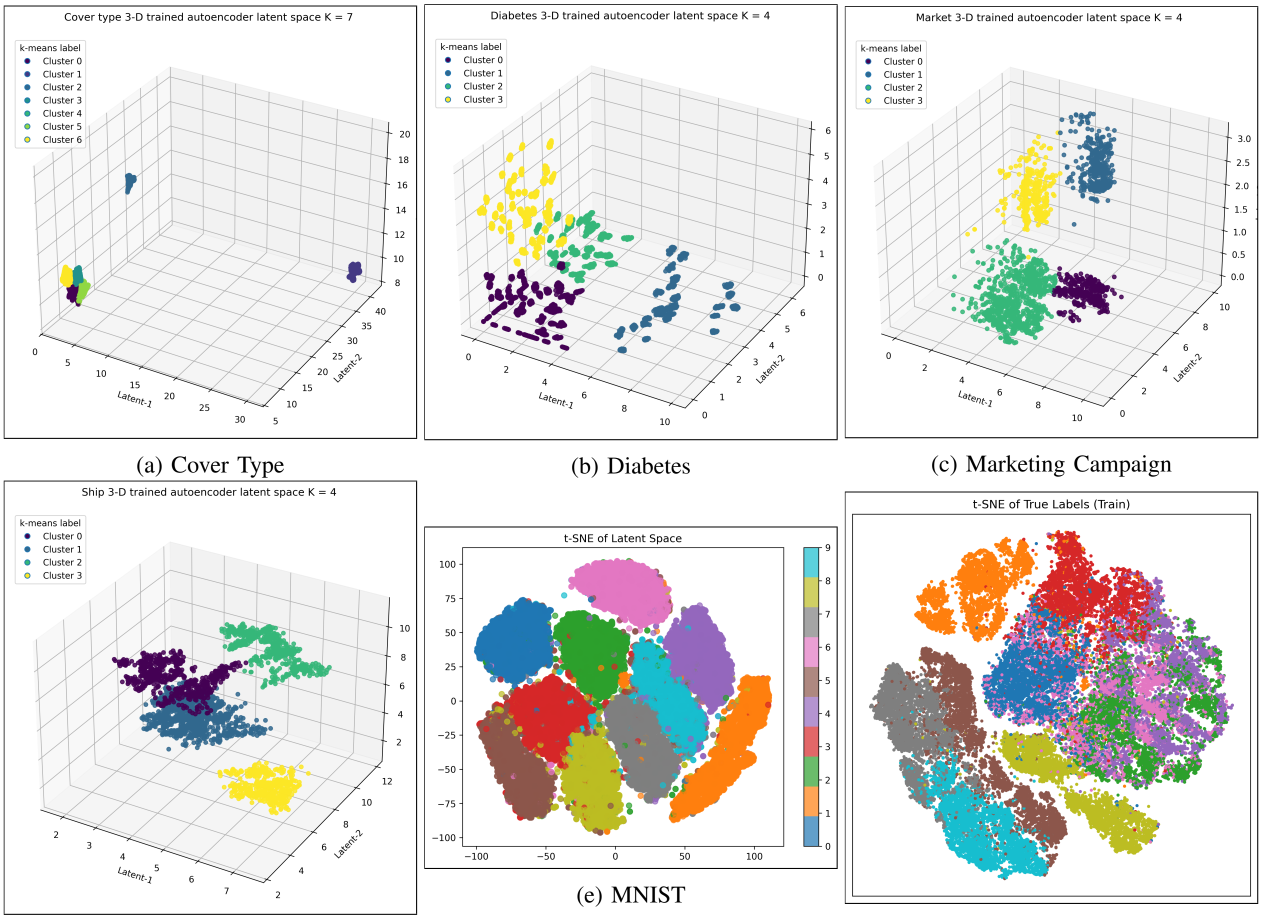
As outlined in the Methodologies section, employing Local Intrinsic Dimensionality (LID) as a proxy for Total Correlation (TC) to learn disentangled representations is not grounded in robust theoretical foundations. Nonetheless, this approach was pursued as an exploratory experiment to empirically examine the nature of the resulting representations. MNIST dataset (26) was used for this experiment. Each image was flattened into a 784-dimensional input vector and passed through an autoencoder with a bottleneck dimension of 10. In an ideally disentangled representation, each latent dimension would independently capture information pertinent to a single digit class. However, as shown in Figure 1a, most latent features disproportionately capture information associated with the digit '1', while exhibiting low mutual information with other labels. Moreover, the uniformity of rows in the mutual information matrix suggests that the latent features are functioning similarly, indicating an absence of feature specialization or disentanglement. This observation is reinforced in Figure 1b, which reveals that each latent feature shares an approximately equal amount of mutual information with label '0', further confirming the lack of disentanglement in the learned representation.

B. Learning cluster-friendly latent space

In this section, we examine two experimental configurations: one where the Itakura–Saito (IS) divergence loss, incorporating Local Intrinsic Dimensionality (LID), fully replaces the Kullback–Leibler (KL) divergence loss; and another where both IS and KL losses are jointly applied alongside the existing reconstruction and K-means losses. The motivation behind evaluating these scenarios separately stems from the differing roles each loss plays: IS loss primarily enforces geometric regularization by encouraging tighter, more linearly separable clusters, whereas KL loss acts as a cluster assignment regularizer, preserving the local similarity structure by bringing similar instances closer and distancing dissimilar ones.

To establish a baseline, Principal Component Analysis (PCA) was performed to reduce each dataset to three dimensions, followed by K-means clustering. This approach serves as a classical benchmark for clustering performance after dimensionality reduction. As shown in Figure 2, the resulting clusters appear diffuse and poorly separated, indicating limited clustering efficacy in the PCA-reduced space.

Next, we analyze the latent embeddings obtained from a model trained using IS loss in place of KL loss, in conjunction with reconstruction and K-means clustering losses (see Figure 3). The number of clusters was specified as input.



(a) Cover Type

(b) Diabetes

(c) Marketing Campaign

(d) Ship Performance

(e) MNIST

(f) Fashion MNIST

Fig. 4: Itakura-Saito Divergence loss using estimated LID used with KL divergence loss

The method demonstrated strong performance on the Cover Type and Ship Performance datasets, yielding well-separated, compact clusters. Results on the Diabetes and Marketing Campaign datasets were satisfactory, though less pronounced. For MNIST, due to the high intrinsic dimensionality of the data, reducing the bottleneck to three dimensions caused instability and significant loss. Instead, the model was trained with a bottleneck dimension of 32, and the learned latent representations were visualized in two dimensions using t-SNE. Compared to PCA+K-means, this approach provided a substantial improvement in clustering quality.

Finally, we explore the effect of combining IS loss with KL divergence loss. This composite objective is designed to synergize the strengths of both terms—where K-means and KL loss jointly manage cluster assignment, and IS loss enforces compact and separable cluster geometry. Results are shown in Figure 4. For MNIST and Fashion MNIST, the latent dimensions were again reduced to 32 before applying t-SNE for visualization. The MNIST dataset exhibited a noticeable improvement in cluster separability compared to using IS loss alone. Fashion MNIST also demonstrated clearer cluster boundaries than PCA-based embeddings, although some overlap remained. This could be attributed to the model being trained for only 100 epochs and the lack of a deeper network architecture, which may have enabled direct compression into 2D or 3D latent space as achieved for other datasets.

Performance on the Diabetes dataset (Figure 4b) declined, likely due to the small dataset size. In such scenarios, the IS loss received noisy and unstable targets, resulting in high-variance gradients that overwhelmed the KL signal and destabilized training. For the Cover Type dataset (Figure 4a), while clusters were well-defined, the IS loss overly tightened them. A subset of points with highly similar features and LID values collapsed into four merged clusters, while the remaining clusters remained well-separated—highlighting the sensitivity of IS loss to local feature distributions.

V. Limitations and Future Work

The purpose of this report was to do a theoretical and empirical investigation into using LID to produce disentangled or cluster friendly latent spaces. The results obtained open up a multitude of possibilities for future work.

- The attempt to achieve disentanglement using LID as a proxy for Total Correlation (TC) was unsuccessful, both theoretically and empirically. Around any given point x, LID estimates the minimal number d of orthogonal directions required to span its local neighborhood. In an ideally disentangled latent space, one expects the global intrinsic dimensionality (ID) to match the number of latent factors, while the local ID should approximate one. When two or more latent coordinates are entangled, movement along one axis influences others, resulting in a local ID greater than one. While TC fails to capture this reliably—particularly due to divergence as neighborhood radii shrink—this behavior may still be leveraged in alternative metrics that are both expressible in terms of LID and stable in the limit of vanishing neighborhood size.

- Disentanglement in representation learning is traditionally explored within a single high-dimensional modality. However, extending it to multi-modal data offers valuable opportunities for uncovering cross-modal structures. By utilizing multiple encoders and decoders, it is possible to independently process and reconstruct distinct modalities while learning a shared latent representation. This facilitates structured disentanglement across modalities, such as correlating specific visual regions with associated textual descriptions in vision-language models. Such architectures have the potential to improve interpretability and enhance cross-modal reasoning in deep neural systems.
- As demonstrated, LID estimation becomes unreliable in small datasets. In high ambient dimensions, the distinction between nearest and farthest neighbor distances diminishes, reducing the effectiveness of distance-based ratios such as r_2/r_1. Consequently, both the Two-NN and MLE-based LID estimators exhibit numerical instability. A single global ID is often inadequate for data distributed across multiple sub-manifolds or manifolds exhibiting nested structures. In such cases, estimators yield inconsistent results depending on the neighborhood size (radius or k). The diabetes dataset (Figure 4b) illustrated this instability, where erroneous LID estimates compromised the learning signal. As such, LID should not be included as an optimized loss term in these settings.
- The methods explored in this report did not directly influence cluster membership assignments but rather served to refine the geometry of clusters that had already been formed. This principle of post hoc cluster refinement can be applied to other clustering algorithms to yield tighter, smoother, and more separable clusters, thereby enhancing clustering quality without altering initial label assignments.

VI. Conclusion

This report explored the theoretical foundations and empirical effectiveness of utilizing Local Intrinsic Dimensionality (LID) to achieve disentangled and cluster-friendly latent representations in autoencoders. The results indicate that directly substituting Total Correlation with LID does not provide theoretically sound or empirically effective disentanglement. Nevertheless, incorporating Itakura–Saito Divergence alongside Kullback–Leibler Divergence has shown promising results in learning latent spaces conducive to clustering, notably improving cluster compactness and separability.

Through comprehensive empirical analyses across diverse datasets, it was demonstrated that careful design of loss functions and architecture modifications significantly impacts the quality of latent space representations. However, challenges remain, especially in handling small datasets or those with inherently high dimensionality. Thus, future studies should focus on robust dimensionality estimation methods, multimodal data integration, and further refinement of clustering methods to fully exploit the capabilities of latent representation learning.

VII. Code and Dataset availability

Project repo: repo_link
Cover type: link_cover
Market: link_market
Ship Performance: link_ship
Diabetes: link_diabetes
MNIST: (26)
Fashion MNIST: (27)

References

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